INCH-POUND

MIL-A-25915A(USAF)
AMENDMENT 2
28 October 1988
SUPERSEDING
AMENDMENT 1
3 November 1983

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT, PILOT'S
WARNING, TYPE MA-2

This amendment forms a part of MIL-A-25915A(USAF), dated 30 April 1980, and is approved for use within the Department of the Air Force and is available for use by all Deparments and Agencies of the Department of Defense.

PAGE 1

* Delete document title and add new title "ACCELEROMETER, MECHANICAL, AIRCRAFT, PILOT'S WARNING, TYPE MA-2".

* 2.1 Delete " title and first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3)."

* Under SPECIFICATION, FEDERAL, delete "PPP-B-601".

* Under SPECIFICATION, MILITARY, delete "MIL-E-5272" and "MIL-S-6872".

Delete address on lines 3 and 4 of beneficial comments block and add: "Oklahoma City Air Logistics Center/MMEDOA, Tinker AFB OK 73145-5990".

PAGE 2

* Under STANDARDS, MILITARY, delete "MIL-STD-143", "MIL-STD-794" and add: "MIL-STD-810 Environmental Test Methods", "MIL-STD-970 Standards & Specification, Order Of Preference For The Selection Of", "DOD-STD-1866 Soldering Process General (Non electrical)" and "MIL-STD-2073-1 DOD Material Procedures For Development And Application Of Packaging Requirement."

* Delete the "parenthetical source statement that follows the listing of Government specifications and standards" and add: "(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)"

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

* Add: "2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings; and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitatin."

AIR FORCE DRAWING 8346561
(Copies of Air Force drawing are available through OC-ALC/MMEDOE TINKER AFB OK 73145-5990.)

* Add: "2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issue of documents not listed in the DODISS are the issue of the documents cited in the solicitation (6.3).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* 3.2 Line 3 delete "MIL-STD-143" and add: "MIL-STD-970".

PAGE 4

* 3.5.g Delete paragraph entirety.

* 3.5.i Line 2 delete "500" and add: "8.3". Line 3 Delete "3,000 cpm" and add: "50Hz".

* 3.5.j Line 2 delete "0.2g" and add: "0.25g"

PAGE 6

* 3.5.k Delete paragraph entirety and add new "3.5k Static scale error. Static scale error shall not exceed ±0.2g, when tested as specified."

* 3.5. Delete paragraph entirety and add new "3.5. "Dynamic scale error. Dynamic scale error shall not exceed ±0.25g at any specified test point.

* 3.5.o Line 3 delete "cps" and add: "Hz".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

PAGE 9

* Delete "3.11" entirety.

* Delete "3.11.1" entirety.

PAGE 10

* 3.13 Line 2 delete "MIL-S-6872" and add: "DOD-STD-1866".

PAGE 12

* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

* 4.3.1 Delete entirety.

PAGE 14

* 4.4.2 Delete entirety.

* 4.4.3 Line 2 delete "500", "3,000 cpm" and add: "8.3"; "50 Hz".

* 4.4.4 Line 8 delete "22" and add: "9.375".

PAGE 15

* 4.5.2 Delete the last "five sentence".

* 4.5.3 Delete paragraph entirety and add new"4.5.3 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and centrifuge rotor per Air Force Drawing 8346561 and tested for dynamic scale error and friction for acceleration of -2g to 8g at 1g increments for a total of 10 test points. For the first test point and without instrument vibration, the centrifuge shall be brought up to, but not exceeding, the rotation rate equivalent to the "g" level being tested. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. After re-zeroing the memory pointer, the test shall be repeated except instrument vibration shall be applied utilizing an electric motor and an attached counterweight as defined by Air Force Drawing 8346561. The centrifuge shall again be reduced to zero and the accelerometer reading taken from the memory pointer. Again, re-zero the memory pointer. The reading with and without vibration shall not differ from the centrifuge input "g" reading by more than $\pm$.25g. This procedure shall be repeated for the remaining 9 test points. The centrifuge rate need not be returned to zero and the memory pointer reset, if capability exists to read the instruments without reducing the centrifuge's rotational rate

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

PAGE 16

* 4.5.4 Delete paragraph entirety.

* 4.5.5 Line 7 delete "cps" and add: "Hz".

* 4.5.5.1 Line 4 delete "cps" and add: "Hz".

4.5.7 Line 2 delete "procedure I of MIL-E-5272 for a period of 4 hours." and add: "method 502.1, procedure I of MIL-STD-810 for a period of 4 hours, except that the storage temperature of step 2 shall be -54°C (-65°F) and also, delete steps 4 through 7."

PAGE 17

* Add: "(Hz) after ACCELERATION

PAGE 18

4.5.8 Line 2 delete "procedure II of MIL-E-5272 for a period of 4 hours." and add: "method 501.1 procedure I of MIL-STD-810 for 4 hours except delete steps 4 through 7."

4.5.10 Line 10 delete "500 to 3,000 cycles per minute" and add: "8.3 to 50 Hz". Line 12 delete "500 and 3000 cpm" and add: "8.3 to 50 Hz". Line 14 delete "1,800 cpm" and add: "30Hz". Line 17 delete "1,200 cpm" and add: "20 Hz". Line 18 delete "1,200 cpm" and add: "20 Hz".

4.5.11 Delete paragraph entirety and add new: 4.5.11 Vibration endurance. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +1 at 270°. The instrument shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plan inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a period of 50 hours. Upon completion of the test period, the accelerometer shall be subjected to the test specified in 4.5.3. The dynamic scale error shall not exceed 0.2g. The accelerometer shall not be damaged as a result of this test.

PAGE 19

4.5.12 Line 3 delete "4.5.11.1" and add: "4.5.12.1".

4.5.14 Lines 2 and 3 delete "low temperature test, procedure VI of MIL-E-5272 and an altitude of 70,000 feet (1.31 inches Hg)" and add: "low temperature test, method 502.1 procedure I of MIL-STD-810, and an altitude of 70,000 feet (1.31 inches Hg) except that the storage temperature for step 2 shall be -62°C (80°F) and the stabilization period shall be 72 hours. The operating temperature of step 4 shall be 24 hours."

4.5.15 Lines 2 and 3 delete "procedure II of MIL-E-5272" and add: "method 501.1, procedure I, of MIL-STD-810."

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

4.5.17 Line 2 delete "humidity test in accordance with procedure I of MIL-E-5272" and add: humidity test in accordance with method 507.1, procedure I of MIL-STD-810, except for the following changes: Step 2, delete 65°C (149°F) and add 71°C (160°F); Step 4, delete 85 percent and add: 95 percent and delete 30°C (86°F) and add 20 to 38°C (68 to 100°F) Step 5, delete the last sentence; Step 6, delete and add "At the end of the tenth cycle, the equipment shall be returned to standard conditions."

4.5.18 Lines 2 and 3 delete "procedure I of MIL-E-5272." and add: "method 508.2 of MIL-STD-810 for a period of 28 days. Evidence of fungus growth which would significantly deteriorate accelerometer performance shall be cause for rejection."

4.5.19 Line 2 delete "procedure I of MIL-E-5272" and add: "method 509.1 , procedure I of MIL-STD-810".

4.7 through 4.7.2: Delete entirely and add new paragraph as follows: "4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

* Delete Section 5 in its entirety and add:

"5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.3).

5.1.1 Level A.

5.1.1.1 Cleaning. Accelerometers shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Accelerometers shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each accelerometer shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each accelerometer shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit containers of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the accelerometer when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each accelerometer shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and phhysical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of accelerometers shall be in accordance with ASTM D3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.3).

5.2.1 Level A. Accelerometers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Accelerometers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged accelerometer shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.3), interior and exterior containers shall be marked in accordance with MIL-STD-129."

PAGE 22

Delete "6.2.1" entirety.

Delete "6.2.2" entirety.

* 6.3 Delete "title and first sentence" and add new: "Acquisition requirements. Acquisition documents must specify the following:".

PAGE 23

6.4 Line 4 delete period after "10/4" and add "and NAT-STD-3330."

*Delete "6.6" entirety.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodian:
Air Force -99

Preparing activity:
Air Force -71

Project No.
6610-F271

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.